

# Lecture Title Here

Course Name — Lecture N

Your Name

Department of Economics, NTNU

May 13, 2026



Norwegian University of  
Science and Technology

2026-05-13

Lecture Title

Lecture Title Here  
Course Name — Lecture N

Your Name

Department of Economics, NTNU  
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Welcome everyone to today's lecture.

- Introduce yourself: name, office hours, email.
- Remind students this is Lecture N of the course.
- Mention the recording policy (if applicable).
- Ask: “Has everyone accessed the course page?”

**Timing:** spend no more than 2 minutes here.

- 1 Introduction
  - Objectives
  - Motivation
- 2 Key Concepts
  - Definitions
  - Theory
- 3 Examples
  - Worked Example
  - Application
- 4 Summary
  - Key Takeaways
- 5 References
  - Bibliography

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Lecture Title

└ Outline

Walk through the outline briefly.

- 1. Introduction — why we care about this topic.
- 2. Key Concepts — the theoretical backbone.
- 3. Examples — hands-on application.
- 4. Summary — what to remember for the exam.

Tell students: “Feel free to stop me at any point with questions.”

| Outline |                                                                                             |
|---------|---------------------------------------------------------------------------------------------|
| 1       | Introduction <ul style="list-style-type: none"><li>Objectives</li><li>Motivation</li></ul>  |
| 2       | Key Concepts <ul style="list-style-type: none"><li>Definitions</li><li>Theory</li></ul>     |
| 3       | Examples <ul style="list-style-type: none"><li>Worked Example</li><li>Application</li></ul> |
| 4       | Summary <ul style="list-style-type: none"><li>Key Takeaways</li></ul>                       |
| 5       | References <ul style="list-style-type: none"><li>Bibliography</li></ul>                     |

# Introduction

## Section 1

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Lecture Title  
└ Introduction

Introduction

Section 1

# Lecture Objectives

By the end of this lecture, you should be able to:

- Objective 1: describe the concept
- Objective 2: apply the method
- Objective 3: interpret the results

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Lecture Title  
└─ Introduction  
    └─ Objectives  
        └─ Lecture Objectives

Lecture Objectives

By the end of this lecture, you should be able to:

- Objective 1: describe the concept
- Objective 2: apply the method
- Objective 3: interpret the results

Read each objective aloud and connect it to an assessment:

- Objective 1 → tested in Problem Set 3, Q2.
- Objective 2 → needed for the group project.
- Objective 3 → will appear in the midterm.

**Key message:** by the end of today, you should be comfortable setting up the model and interpreting output — not just memorising formulas.

- Why this topic matters
- Connection to previous lectures
- Real-world relevance

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Lecture Title

- └ Introduction
  - └ Motivation
    - └ Motivation

Motivate the topic with a concrete story:  
“In 2023, researchers at the Norwegian central bank used exactly this method to evaluate the effect of interest rate changes on household spending. That’s the kind of question we’ll learn to answer today.”  
  
**Link to last lecture:** Recall the reduced-form approach from Lecture N–1. Today we move to a structural model — ask students what they think the difference is before explaining.

Motivation

- Why this topic matters
- Connection to previous lectures
- Real-world relevance

# Key Concepts

Section 2

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Lecture Title  
└─ Key Concepts

Key Concepts

Section 2

Definition 1

A formal definition of the key concept goes here.

Definition 2

Another important definition for this lecture.

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- Lecture Title
  - Key Concepts
    - Definitions
      - Core Definitions

Read each definition slowly and pause after each one.

**Definition 1:** Ask the class: “Has anyone encountered this term before? In what context?” — use their answers to anchor the formal statement in intuition.

**Definition 2:** Emphasise the subtle distinction from Definition 1. Common student confusion: they mix up the two because both involve conditional expectations.

*Tip:* Write both definitions side-by-side on the whiteboard and circle where they differ.

Core Definitions

Definition 1

A formal definition of the key concept goes here.

Definition 2

Another important definition for this lecture.

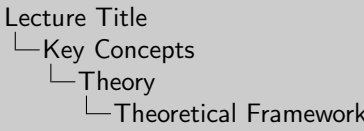
- Main theoretical result or proposition
- Key assumptions and conditions
- Intuition behind the result

The central equation:

$$Y_{i,t} = \alpha + \beta X_{i,t} + \gamma Z_{i,t} + \epsilon_{i,t}$$

(1)

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Walk through the equation term by term:

- $Y_{i,t}$  Outcome we want to explain (e.g. household spending).
- $X_{i,t}$  Treatment / policy variable (e.g. interest rate change).
- $Z_{i,t}$  Controls — list 2–3 concrete examples.
- $\epsilon_{i,t}$  Everything we cannot observe.

**Identification assumption:**  $\mathbb{E}[\epsilon_{i,t} \mid X_{i,t}, Z_{i,t}] = 0$ . Ask: “When might this fail?” — guide students toward omitted variable bias and reverse causality.

**Timing:** ~5 minutes on this slide.

Theoretical Framework

- Main theoretical result or proposition
- Key assumptions and conditions
- Intuition behind the result

The central equation:

$$Y_{i,t} = \alpha + \beta X_{i,t} + \gamma Z_{i,t} + \epsilon_{i,t}$$

(1)



# Examples

Section 3

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Lecture Title  
└─ Examples

Examples

Section 3

# Worked Example

## Example 1

Consider the case where  $X = 1$  and  $Z = 0$ . Then the model reduces to:

$$Y_{i,t} = \alpha + \beta + \epsilon_{i,t}$$

Interpretation:  $\beta$  captures the treatment effect directly.

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└ Examples

└ Worked Example

└ Worked Example

Example 1

Consider the case where  $X = 1$  and  $Z = 0$ . Then the model reduces to:

$$Y_{i,t} = \alpha + \beta + \epsilon_{i,t}$$

Interpretation:  $\beta$  captures the treatment effect directly.

Work through the algebra on the board:

1. Start with the full model:  $Y = \alpha + \beta X + \gamma Z + \epsilon$ .
2. Plug in  $X = 1, Z = 0$  — show cancellation explicitly.
3. Result:  $Y = \alpha + \beta + \epsilon$ .
4. Interpretation:  $\beta$  is the treatment effect for treated units.

**Common mistake:** students forget that  $\alpha$  is still present and try to interpret  $\beta$  as the level of  $Y$ .

Ask: “What would  $\beta < 0$  mean in our central bank example?”

| Variable  | Description         | Source     |
|-----------|---------------------|------------|
| $Y_{i,t}$ | Outcome variable    | Database A |
| $X_{i,t}$ | Treatment indicator | Database B |
| $Z_{i,t}$ | Control vector      | Database C |

Table 1: Variable definitions and data sources

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|                               |
|-------------------------------|
| Lecture Title                 |
| └─ Examples                   |
| └─ Application                |
| └─ Application: Data Overview |

Describe the data:

- **Database A** (outcome): annual panel, 2000–2022,  $N \approx 5,000$  observations. Mention any cleaning steps.
- **Database B** (treatment): binary indicator coded from policy announcements. Stress the exact timing definition.
- **Database C** (controls): standard macro controls — GDP growth, unemployment, CPI.

**Exclusions:** dropped municipalities with  $<100$  observations. Explain why.  
If students ask about data access — point them to the course GitHub repository where a sample extract is available.

Application: Data Overview

| Variable  | Description         | Source     |
|-----------|---------------------|------------|
| $Y_{i,t}$ | Outcome variable    | Database A |
| $X_{i,t}$ | Treatment indicator | Database B |
| $Z_{i,t}$ | Control vector      | Database C |

Table 1: Variable definitions and data sources

# Summary

Section 4

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Lecture Title  
└ Summary

Summary

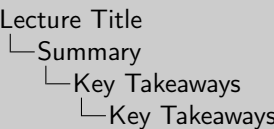
Section 4

# Key Takeaways

- ① Takeaway 1: summary of the main concept
- ② Takeaway 2: summary of the key method
- ③ Takeaway 3: practical implications

**Next lecture:** Topic of the following session.

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Recap — repeat each takeaway in one sentence:

1. We defined the key concept and its formal conditions.
2. We learned how to set up and estimate the baseline model.
3. We saw that the results have direct policy implications.

**Preview next lecture:** “Next week we extend this to a difference-in-differences design. Read Chapter 5 of the textbook before class.”

Remind students of the problem set deadline (Friday 23:59).

- Takeaway 1: summary of the main concept
- Takeaway 2: summary of the key method
- Takeaway 3: practical implications

**Next lecture:** Topic of the following session.

# References

Section 5

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References

References

Section 5

[1]

Author, F. and Author, S. *Title of the Article*. Journal Name, 2024.

[2]

Author, T. *Textbook Title*, 3rd ed. Publisher, 2023.

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└─References

└─Bibliography

└─References

Go through the reference list:

- **Required:** Article 1 — read Sections 1–3 carefully; Section 4 (robustness) is optional but recommended.
- **Background:** Textbook Chapter 4 covers the same material at a slower pace — good for students who want more intuition before the math.

Remind students: “These readings will be assumed knowledge in the next lecture. I will cold-call on the key results.”

# Thank You!

Your Name

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Lecture Title  
└─References  
└─Bibliography

Thank You!

Your Name  
`your.email@ntnu.no`  
Department of Economics, NTNU  


## Q&A time.

- “Any questions about today’s material?”
- If silence: prompt with “What part of the identification argument felt least convincing?”
- Remind: office hours are Wednesdays 14:00–15:30, Room A301.
- Problem Set 3 is due Friday — encourage starting early.

Thank the students for their attention.



Items appear one at a time (overlay specification):

- First point (appears alone, highlighted)

Items appear one at a time (overlay specification):

- First point (appears alone, highlighted)
- Second point (added next)

Items appear one at a time (overlay specification):

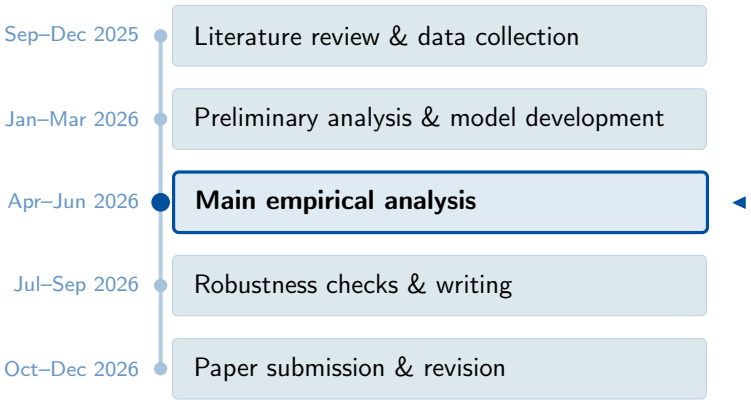
- First point (appears alone, highlighted)
- Second point (added next)
- **Third point (added last)**

Standard Block  
Used for general information or definitions.

Alert Block  
Used for warnings or important notes.

Example Block  
Used for examples or illustrations.

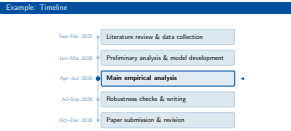
Example: Timeline



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Lecture Title

- └ Template Examples
- └ Example: Timeline



| Phase              | Period       | Deliverable        |
|--------------------|--------------|--------------------|
| Literature review  | Sep–Dec 2025 | Survey paper draft |
| Data collection    | Jan–Feb 2026 | Clean dataset      |
| Model development  | Feb–Mar 2026 | Working model      |
| Empirical analysis | Apr–Jun 2026 | Results tables     |

Table 2: Project milestones

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- Lecture Title
  - Template Examples
    - Example: Table

Example: Table

| Phase              | Period       | Deliverable        |
|--------------------|--------------|--------------------|
| Literature review  | Sep–Dec 2025 | Survey paper draft |
| Data collection    | Jan–Feb 2026 | Clean dataset      |
| Model development  | Feb–Mar 2026 | Working model      |
| Empirical analysis | Apr–Jun 2026 | Results tables     |

Table 2: Project milestones

The baseline regression:

$$Y_{i,t} = \alpha + \beta X_{i,t} + \gamma Z_{i,t} + \epsilon_{i,t}$$

(2)

where  $Y_{i,t}$  is the dependent variable,  $X_{i,t}$  is the treatment, and  $Z_{i,t}$  is a vector of controls.

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└ Template Examples

└ Example: Equation

Example: Equation

The baseline regression:

$$Y_{i,t} = \alpha + \beta X_{i,t} + \gamma Z_{i,t} + \epsilon_{i,t}$$

(2)

where  $Y_{i,t}$  is the dependent variable,  $X_{i,t}$  is the treatment, and  $Z_{i,t}$  is a vector of controls.

Left column

- Point A
- Point B

Right column

- Point C
- Point D

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- └ Template Examples
  - └ Example: Two Columns

Example: Two Columns

Left column

- Point A
- Point B

Right column

- Point C
- Point D